DAY 8 – CAPTURE ANALYSIS

Cybersecurity Internship | Network Traffic Investigation Report

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Task: Analyse the supplied packet capture file (day8_capture.pcap) using Wireshark.

Tool: Wireshark

Purpose

This report documents a basic network traffic investigation. It reviews the packet capture, identifies exposed HTTP credentials, examines a SYN scan, and highlights a repeated outbound connection pattern that may indicate command-and-control (C2) beaconing.

Important: The IP addresses, usernames, password, ports, and timing below are observations from the supplied training capture. They should be treated as lab evidence, not proof of activity on a real production network.

1. Executive Summary

Area	Observation	Assessment
Capture overview	97 packets; 6 unique IPv4 addresses; TCP and UDP traffic observed.	Small training capture suitable for manual inspection.
Finding 1 – HTTP POST	10.0.0.15 sent a POST request to 10.0.0.20 over TCP port 80. Form data exposed username admin and password Tekl00my@123.	Credentials are visible because HTTP is unencrypted.
Finding 2 – SYN scan	10.0.0.66 sent SYN probes to 13 ports on 10.0.0.20. SYN-ACK responses indicate ports 22 and 80 were open.	Reconnaissance / port scanning behaviour.
Finding 3 – Beacon pattern	10.0.0.15 repeatedly communicated with 185.220.101.47:8080, with approximately 15.01 seconds between messages.	Suspicious periodic traffic; investigate as possible C2 beaconing.

2. Warm-up – Capture Overview

Check	Result	Meaning
Total packets	97	Total number of captured frames in this file.
IPv4 endpoints	6 unique IPv4 addresses	The capture contains six distinct IPv4 hosts.
Protocols observed	Ethernet II, IPv4, TCP, UDP, DNS; HTTP application traffic is also present.	TCP is the dominant transport protocol (95 TCP packets and 2 UDP packets).

Wireshark navigation: open the capture, read the packet count in the bottom status bar, then use Statistics → Protocol Hierarchy and Statistics → Endpoints. In Endpoints, select IPv4 to count the unique IPv4 hosts.

3. Finding 1 – Exposed Credentials in HTTP POST

Display filter: **http.request.method == "POST"**

Field	Observed value
Client / source IP	10.0.0.15
Server / destination IP	10.0.0.20
Destination port / protocol	TCP port 80 / HTTP
HTTP form username	admin
HTTP form password	Tekl00my@123
Raw form body	username=admin&password=Tekl00my%40123

How to verify in Wireshark: apply the POST display filter, select the matching packet, right-click it, choose Follow → TCP Stream, and inspect the HTTP request body. The encoded text %40 represents the @ character.

Why this is a security issue: HTTP sends application data without TLS encryption. Anyone able to capture the traffic on the network path may be able to read the submitted credentials.

Recommended fix: use HTTPS with correctly configured TLS, redirect HTTP to HTTPS, enable HSTS where appropriate, and reset any credentials exposed in the capture. Avoid reusing the exposed password.

4. Finding 2 – SYN Port Scan

SYN probe filter: **tcp.flags.syn == 1 && tcp.flags.ack == 0**
SYN-ACK response filter: **tcp.flags.syn == 1 && tcp.flags.ack == 1**

Field	Observed value
Scanner / source IP	10.0.0.66
Target / destination IP	10.0.0.20
Destination ports probed	21, 22, 23, 25, 80, 110, 139, 143, 443, 445, 3306, 3389, 8080 (13 ports)
Ports with SYN-ACK responses	22 and 80

Port	Common service	Typical purpose
21	FTP	File transfer (often unencrypted).
22	SSH	Secure remote login and administration.
23	Telnet	Remote terminal access (unencrypted).
25	SMTP	Email transfer between mail servers.
80	HTTP	Web traffic without TLS encryption.
110	POP3	Email retrieval.
139	NetBIOS	Legacy Windows file/printer sharing services.
143	IMAP	Email access and synchronization.
443	HTTPS	Web traffic protected by TLS.
445	SMB	Windows file and printer sharing.
3306	MySQL	MySQL database service.
3389	RDP	Remote Desktop access.
8080	Alternative HTTP / web proxy	Common alternate web application or proxy port.

Interpretation: a SYN scan sends initial TCP connection requests to discover listening services. A SYN-ACK response normally indicates that the destination port is open; a reset response commonly indicates a closed port. The capture shows SYN-ACK responses for ports 22 (SSH) and 80 (HTTP).

5. Finding 3 – Possible C2 Beaconing

Field	Observed value
Internal host	10.0.0.15
External destination	185.220.101.47
Destination port	8080
Observed pattern	Repeated PING messages with PONG replies; 20 PING messages observed.
Approximate interval	15.01 seconds between repeated messages.
Comparison traffic	Other HTTPS connections to 142.250.182.4 on port 443 appear as a short burst rather than the same regular repeating pattern.

Why it is suspicious: malware may periodically contact a remote command-and-control server to check for instructions or report status. A repeated connection at a nearly fixed interval can be a beaconing indicator. However, periodic traffic alone does not prove malware; the destination, payload, host activity, and endpoint logs should also be examined.

Recommended investigation: identify the process on 10.0.0.15 that generated the traffic, review DNS and firewall/proxy logs, inspect the destination and payload, preserve the PCAP and endpoint evidence, and block the destination only under the organisation's incident-response process.

6. Mini Incident Report (Who / What / Fix)

Who	What happened	Recommended response / Fix
10.0.0.15	Sent HTTP POST credentials to 10.0.0.20 and showed repeated periodic traffic to 185.220.101.47:8080.	Isolate the host for investigation; preserve evidence; identify the process responsible; reset exposed credentials; enforce HTTPS/TLS.
10.0.0.66	Sent SYN probes to multiple ports on 10.0.0.20.	Confirm whether the scan was authorised. Review host and firewall logs, and restrict unnecessary access.
10.0.0.20	Responded to scan probes on ports 22 and 80 with SYN-ACK packets.	Check whether SSH and HTTP are required; harden SSH and use HTTPS instead of HTTP.

7. Conclusion

The capture contains three notable security observations: credentials transmitted in plaintext over HTTP, a SYN scan from 10.0.0.66 against 10.0.0.20, and periodic outbound communication from 10.0.0.15 to 185.220.101.47 on port 8080. The highest-priority actions are to investigate 10.0.0.15, protect/reset the exposed credentials, verify whether the scan was authorised, and review the repeated external communication as a possible C2 beacon.

8. Screenshot Evidence Checklist

■ Screenshot 1: Wireshark packet list showing the total packet count and opened day8_capture.pcap.

■ Screenshot 2: Statistics → Protocol Hierarchy.

■ Screenshot 3: Statistics → Endpoints with IPv4 selected.

■ Screenshot 4: POST display filter and Follow → TCP Stream showing the request details (mask the password if sharing outside the training submission).

■ Screenshot 5: SYN scan filter showing source 10.0.0.66 and destination 10.0.0.20.

■ Screenshot 6: SYN-ACK filter showing ports 22 and 80.

■ Screenshot 7: Statistics → Conversations → TCP sorted by packet count, with the relevant 10.0.0.15 to 185.220.101.47 conversation visible.

Submission reminder: add your own Wireshark screenshots to this report, save/export the final document as PDF, and upload it to Portal → Day 8 → Upload before the stated deadline.

Prepared as a training report from the supplied Day 8 packet-capture exercise.